

HCSM-26

The Observer Pair $\{W, t\}$

Fixed Locus, Chirality Selection, Uniqueness,
and the Klein Orbit Structure

The Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model assigns the fourteen modes at the self paired eigenvalue of the discrete Laplacian on the periodic 8×8 torus to one Standard Model generation. The assignment of each mode to a specific Standard Model particle, together with the $N/4$ mass label attached to each, is the framework's *empirical thesis* (HCSM-23), structurally parallel to the Standard Model's Yukawa-coupling sector. The substrate *derives* the fourteen modes, their coordinates, their D_4 orbits, the winding generator and its class partition, the processor T and its F/B split, the chiral reflection rs and its fixed locus, and the zero state s . *Given* the empirical identification, the structural relations among the $N/4$ anchors are theorems of the framework.

Among the fourteen modes at $\lambda = 4$, the coordinate pair $\{(2, 2), (6, 6)\}$ is distinguished: it is the fixed locus of the unique chiral reflection of the D_4 point group of the square. Under the identification thesis, this pair is the W boson and the top quark and is called the observer pair $\{W, t\}$. This paper proves eight results about the pair. Two of them are fully derived from the substrate and do not use the $N/4$ anchors: the Fixed-Locus Theorem and the Chirality Selection Theorem. Four of them are theorems *given* the empirical identification: the D_4 -fixed-pair taxonomy, the $(d + 1)/2$ orbit signature, the observer-pair uniqueness theorem, and the swap-pair fingerprint theorem. The remaining results are mixed: each component is labeled according to whether it is derived or anchored. Section 9 establishes a dimensional rigidity statement linking the observer pair to the kL crossover identity of HCSM-51; the statement is two independent constraints on d plus one definitional evaluation, all sharing the same value at $d = 4$. All numerical claims are verified at machine precision.

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1 Introduction

1.1 The observer pair in HCSM

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ [1]. The fourteen modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian are identified with one Standard Model generation. Their D_4 representation theory determines the gauge algebra; their $N/4$ labels determine the mass spectrum; the zero state vector s from the 14×19 vacuum-native measurement matrix determines the flavor structure.

The identification of the fourteen modes with the fourteen Standard Model particles, and the $N/4$ mass label attached to each, is the framework's *empirical thesis* (HCSM-23 [9]). It is not a derived result. The $N/4$ labels are empirical anchors, exactly parallel to the Standard Model's Yukawa couplings: they are the inputs from which particle specific physics is read out, and the structural relations among them are theorems given the anchor.

Throughout this paper:

- The fourteen modes, their coordinates, their D_4 orbits, the winding generator and its class partition, the processor T and its F/B split, the chiral reflection rs and its fixed locus, and the zero state s are *theorems* of the torus.
- The identification of each mode with a specific Standard Model particle, and the $N/4$ label attached to each, are *empirical anchors*.
- The structural relations among the $N/4$ values are *theorems given the anchor*.

Among the fourteen modes, the coordinate pair $\{(2, 2), (6, 6)\}$ carries a distinctive structural status: it is the fixed locus of the unique chiral reflection of the D_4 point group of the square. Under the identification thesis (HCSM-23), this pair is the W boson and the top quark; the pair is called the observer pair $\{W, t\}$. This paper characterizes the pair completely and shows how the residual structure of the fourteen-mode multiplet organizes around it.

1.2 The one-line summary

In HCSM, physics is what the machine cannot cancel. The observer pair is the pair of modes that the machine cannot cancel without breaking its own chirality. The twelve other modes are paired

under the chiral reflection rs ; the pair $\{(2, 2), (6, 6)\}$ is not, because both members are fixed by the reflection that the machine is built to be symmetric under. Under the identification thesis, this pair is the framework's observer pair: the boundary through which the substrate reads itself.

1.3 Derivation status of the results in this paper

The paper proves eight results. They split into three groups according to their derivation status.

1.3.1 Fully derived (no anchor)

These results use only substrate primitives and do not refer to the $N/4$ labels or to the particle identifications:

- (1) **Fixed Locus Theorem** (Section 3). The coordinate pair $\{(2, 2), (6, 6)\}$ is the fixed locus of the diagonal reflection rs of D_4 , and coincides with the zero locus of the zero state vector s and with the $\sigma = 0$ line of the charge theorem. Under the identification thesis, this pair is $\{W, t\}$.
- (2) **Chirality Selection Theorem** (Section 4). The diagonal reflection rs is the unique element of D_4 that both fixes exactly the coordinate pair $\{(2, 2), (6, 6)\}$ and anticommutes with the winding generator H_{wind} .

1.3.2 Theorems given the empirical identification

These results are conditional on the identification thesis (HCSM-23). They are theorems about the structural relations among the $N/4$ anchors:

- (3) **D_4 -Fixed Pair Taxonomy** (Section 5). The fourteen mode multiplet contains exactly three D_4 -involution-fixed two-element coordinate subsets. Given the identification, these are distinguished in every structural feature: winding class, fermion/boson content, σ -line membership, and $N/4$ ratio.
- (4) **$(d + 1)/2$ Orbit Signature** (Section 6). Given the identification, the framework constant $(d + 1)/2 = 5/2$ manifests in each of the three D_4 orbits via a distinct binary operation: as a ratio in $\mathcal{O}_0$, a difference in $\mathcal{O}_1$, and a sum in $\mathcal{O}_2$.
- (5) **Observer-Pair Uniqueness Theorem** (Section 7). Given the identification, $\{W, t\}$ is the unique pair of $N/4$ values satisfying the two numeric constraints $N/4(W) + N/4(t) = (d + 1)/2$ and $N/4(W) - N/4(t) = (d - 1)/2$ simultaneously with the reflection-fixed and $\sigma = 0$ conditions. The uniqueness is a uniqueness within the family of anchors: it is a structural relation among the $N/4$ values, not a derivation of which mode is W or which is t .
- (6) **Swap-Pair Fingerprint Theorem** (Section 8). Given the identification, the orbit-class swap pair $\{\text{dark}, t\}$ is the unique two-element subset of the fourteen-mode multiplet whose four symmetric-function values (sum, difference, product, ratio) are simultaneously all framework constants.

1.3.3 Mixed results

Section 9 establishes a dimensional rigidity statement. Its components have distinct derivation status and are labeled individually: identity 1 is a theorem given the anchor; identity 2 is a derived theorem of HCSM-51; identity 3 is a definitional evaluation at $d = 4$ given the F/B split.

1.4 Cross references to the HCSM paper stack

The paper depends on the following companion papers:

- HCSM-13 [5]: the processor T and the F/B -class correspondence. Used for the class partition, the fermion/boson split, and the exact correspondence $T = I - P_C$.
- HCSM-14 [6]: the winding generator spectrum and class partition.
- HCSM-22 [8]: the zero state.
- HCSM-23 [9]: the identification thesis and the $N/4$ labels.
- HCSM-24 [10]: the charge theorem $q_3 = F(n_1, \sigma)$. Used for the σ -lines and the charge eigenvalues.
- HCSM-27 [11]: the Klein orbit structure. Referenced in Section 10 for the orbit kernel classification.
- HCSM-51 [13]: the kL crossover. Referenced in Section 9 for identity 2 of the dimensional-rigidity statement.
- HCSM-53 [14]: Dark Energy Density. Referenced for the coefficient $(d+1)/2$.

1.5 Organization

Section 2 states the setup. Sections 3 and 4 prove the two fully-derived theorems. Sections 5–8 prove the four theorems given the anchor. Section 9 establishes the dimensional-rigidity statement. Section 10 discusses the structural interpretation. Section 11 catalogs inputs and derived results. Section 12 gives reproducibility notes. Section 13 concludes.

2 Setup

2.1 The substrate and the fourteen-mode multiplet

The substrate is the periodic eight-by-eight torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites, selected by the photon helicity anchor and the equal-entropy-spacing anchor [1, 2]. The discrete five-point Laplacian

$$(\Delta f)(x, y) = 4f(x, y) - f(x+1, y) - f(x-1, y) - f(x, y+1) - f(x, y-1) \quad (1)$$

has spectrum $\lambda(n_1, n_2) = 4 - 2\cos(\pi n_1/4) - 2\cos(\pi n_2/4)$ for $(n_1, n_2) \in \{0, \dots, 7\}^2$. The self-paired eigenvalue $\lambda = 4$ has multiplicity $2(L-1) = 14$ with $L = 8$.

The fourteen modes at $\lambda = 4$ decompose under the D_4 point group of the square into three orbits:

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad \text{size 2,} \quad (2)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad \text{size 8,} \quad (3)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}, \quad \text{size 4.} \quad (4)$$

Following HCSM-04 [3] and the identification thesis of HCSM-23 [9], these modes are assigned to one Standard Model generation. In the standard mode order used throughout HCSM (matching the zero-state vector of Section 2.5):

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t). \quad (5)$$

The mode coordinates in this order are:

$$\begin{array}{llll} \nu : (0, 4), & e : (4, 0), & u : (1, 3), & d : (1, 5), \\ s : (3, 1), & \mu : (3, 7), & \text{dark} : (5, 7), & c : (7, 3), \\ \tau : (7, 5), & b : (5, 1), & W : (2, 2), & Z : (2, 6), \\ H : (6, 2), & t : (6, 6). & & \end{array}$$

2.2 The D_4 action

The D_4 point group is generated by the rotation

$$r : (n_1, n_2) \mapsto (-n_2, n_1) \pmod{8} \quad (6)$$

and the reflection

$$s : (n_1, n_2) \mapsto (n_1, -n_2) \pmod{8}, \quad (7)$$

subject to the presentation $r^4 = s^2 = (rs)^2 = e$. There are five order-two elements (involutions): r^2, s, rs, r^2s, r^3s . Explicitly,

$$r^2 : (n_1, n_2) \mapsto (-n_1, -n_2), \quad (8)$$

$$s : (n_1, n_2) \mapsto (n_1, -n_2), \quad (9)$$

$$rs : (n_1, n_2) \mapsto (n_2, n_1), \quad (10)$$

$$r^2s : (n_1, n_2) \mapsto (-n_1, n_2), \quad (11)$$

$$r^3s : (n_1, n_2) \mapsto (-n_2, -n_1). \quad (12)$$

The reflection rs is called the diagonal reflection because it swaps the two mode coordinates.

2.3 The winding generator and the class partition

The winding generator on the fourteen-mode multiplet is

$$H_{\text{wind}} = i \sin\left(\frac{\pi}{8}M\right), \quad M_{ij} = n_1^{(i)}n_2^{(j)} - n_1^{(j)}n_2^{(i)}, \quad (13)$$

where $(n_1^{(i)}, n_2^{(i)})$ are the momentum labels of mode i . Its spectrum is

$$\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}. \quad (14)$$

The fourteen modes partition into three winding classes by frequency:

$$A = \{(0, 4), (4, 0), (6, 6)\}, \quad (15)$$

$$B = \{(1, 3), (1, 5), (3, 1), (3, 7), (7, 3), (7, 5), (5, 1)\}, \quad (16)$$

$$C = \{(5, 7), (2, 2), (2, 6), (6, 2)\}. \quad (17)$$

See HCSM-14 [6] for the full derivation of the winding spectrum.

2.4 The processor T

The processor T is derived in HCSM-13 [5] as the complement of the class- C projector:

$$T = I - P_C, \quad (18)$$

from which the class partition and the fermion/boson split satisfy the exact correspondence

$$A \cup B = \text{image}(T) = F, \quad C = \ker(T) = B. \quad (19)$$

The fermion sector is $F = \{\nu, e, u, d, s, \mu, c, \tau, b, t\}$ with $|F| = 10$, and the boson sector is $B = \{\text{dark}, W, Z, H\}$ with $|B| = 4$.

The equivalent characterization of F and B in terms of the framework quantity $\delta = N/4 - \text{base}$ (where base is a framework base formula expressed in d and H) is the subject of HCSM-11 [4]:

$$F = \{\delta \neq 0\} \cup \{s = +1\}, \quad B = \{\delta = 0\} \setminus \{s = +1\}. \quad (20)$$

This characterization is a theorem given the empirical identification (HCSM-23): it depends on the $N/4$ anchors through the δ function. The primary characterization $T = I - P_C$ does not use the anchor.

2.5 The zero state

A 14×19 measurement matrix of vacuum-native observables has rank 13 and nullity 1 [8]. Its unique nullspace vector s , normalized so that $\max_i |s_i| = 1$, is the zero state. In the standard mode order (5):

$$\begin{aligned} s = (& -0.105198199128, +0.105198199128, +0.937726561300, \\ & -0.192806690737, -0.937726561300, +0.658283218429, \\ & -1.000000000000, -0.658283218429, +1.000000000000, \\ & +0.192806690737, 0.000000000000, -0.329141609215, \\ & +0.329141609215, 0.000000000000). \end{aligned}$$

The two coordinates with $s = 0$ exactly are (2, 2) and (6, 6). Under the identification thesis (HCSM-23), these coordinates correspond to the W boson and the top quark respectively. The zero-state statement at the coordinate level is derived; the naming of the coordinates is part of the anchor.

2.6 Charge, $N/4$ labels, and the σ -lines

The charge theorem (HCSM-24 [10]) states that the electric charge in units of $e/3$ is a bijection

$$q_3 = F(n_1, \sigma), \quad n_1 \in \mathbb{Z}_8, \quad \sigma = \text{sign}(s) \in \{-1, 0, +1\}, \quad (21)$$

from the fourteen realized pairs to the five Standard Model charge eigenvalues $\{-3, -1, 0, +2, +3\}$. The three σ -lines are

$$\sigma = -1 : \{\nu, d, s, \text{dark}, c, Z\}, \quad \text{charge sum } 0, \quad (22)$$

$$\sigma = 0 : \{(2, 2), (6, 6)\}, \quad \text{charge sum } +5, \quad (23)$$

$$\sigma = +1 : \{e, \mu, \tau, b, H, u\}, \quad \text{charge sum } -8. \quad (24)$$

The grand total is $-3 = -1e$.

The $N/4$ labels are the framework's empirical anchors for the particle spectrum (HCSM-23 [9]). In the standard mode order:

$$(N/4)_i = (60, 24, 21, 20, 14, 14, 12, 9.5, 9, 7.5, 2, 2, 1, 0.5). \quad (25)$$

The labels are treated as inputs throughout this paper. The structural relations among them are the theorems.

2.7 Notation

Throughout the paper:

- $\mathcal{M}$ denotes the fourteen-mode multiplet at $\lambda = 4$.
- $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$ denote the three D_4 orbits.
- A, B, C denote the three winding classes.
- F, B denote the fermion and boson sectors (image and kernel of T).
- $\sigma = \text{sign}(s)$ denotes the sign of the zero-state amplitude.
- $N/4$ denotes the mass-ordering label of a mode (an empirical anchor).
- $d = 4$ denotes the spacetime dimension.
- $H = 2^d = 16$ denotes the hidden-sector dimension parameter.
- The coordinate pair $\{(2, 2), (6, 6)\}$ is the derived fixed locus of rs ; under the identification thesis it is $\{W, t\}$.

3 The Fixed Locus Theorem

This section proves that the coordinate pair $\{(2, 2), (6, 6)\}$ is the fixed locus of the diagonal reflection rs . The result is derived from the substrate; the naming of the pair as $\{W, t\}$ applies the identification thesis (HCSM-23).

3.1 The fixed set of rs

Lemma 1 (Fixed set of rs). *The diagonal reflection $rs : (n_1, n_2) \mapsto (n_2, n_1)$ acts on the fourteen-mode multiplet with fixed set exactly*

$$\text{Fix}(rs) = \{(2, 2), (6, 6)\}. \quad (26)$$

Proof. A mode (n_1, n_2) is fixed by rs if and only if $n_1 = n_2 \pmod{8}$. Among the fourteen modes at $\lambda = 4$, the diagonal pairs (n, n) with $n \in \{0, \dots, 7\}$ are $(0, 0), (1, 1), \dots, (7, 7)$. Of these, only $(2, 2)$ and $(6, 6)$ satisfy the eigenvalue equation $\lambda = 4$, i.e., $\cos(\pi n/4) = 0$, which requires $n \in \{2, 6\}$. Hence the fixed set is exactly $\{(2, 2), (6, 6)\}$. $\square$

3.2 The zero state vector is odd under rs

Lemma 2 (Oddness of s under rs). *The zero-state vector s satisfies $s(rs(m)) = -s(m)$ for every mode m in the fourteen-mode multiplet.*

Proof. Direct computation on the explicit vector of Section 2.5. For example, $s(\nu) = -0.105198199128$, $s(e) = +0.105198199128$, and $rs(\nu) = e$; $s(u) = +0.937726561300$, $s(s) = -0.937726561300$, and $rs(u) = s$. In every case, $s(rs(m)) = -s(m)$. $\square$

Corollary 3 (Vanishing on the fixed set). *The zero-state vector satisfies $s((2, 2)) = s((6, 6)) = 0$ exactly.*

Proof. By Lemma 1, $rs((2, 2)) = (2, 2)$ and $rs((6, 6)) = (6, 6)$. By Lemma 2, $s(rs((2, 2))) = -s((2, 2))$, hence $s((2, 2)) = -s((2, 2))$, so $s((2, 2)) = 0$. The same argument gives $s((6, 6)) = 0$. $\square$

3.3 The fixed locus and the σ -lines

Theorem 4 (Fixed Locus Theorem). *The following three characterizations of the coordinate pair $\{(2, 2), (6, 6)\}$ coincide:*

$$\{(2, 2), (6, 6)\} = \text{Fix}(rs) = \{m \in \mathcal{M} : s(m) = 0\} = \{\sigma = 0 \text{ line}\}. \quad (27)$$

Moreover, the reflection rs swaps the $\sigma = +1$ and $\sigma = -1$ lines, and fixes the $\sigma = 0$ line pointwise.

Proof. The first equality is Lemma 1. The second is Corollary 3. For the third, by definition $\sigma(m) = \text{sign}(s(m))$, so $\sigma(m) = 0$ iff $s(m) = 0$; this gives $\{m : s(m) = 0\} = \{\sigma = 0 \text{ line}\}$.

For the swap property: by Lemma 2, $s(rs(m)) = -s(m)$ for all m . Therefore $\sigma(rs(m)) = \text{sign}(s(rs(m))) = \text{sign}(-s(m)) = -\sigma(m)$ whenever $\sigma(m) \neq 0$. On the fixed set, $\sigma((2, 2)) = \sigma((6, 6)) = 0$, so the $\sigma = 0$ line is fixed pointwise. $\square$

Remark 5 (Application of the identification thesis). Theorem 4 is a derived result about the coordinate pair $\{(2, 2), (6, 6)\}$. It does not refer to W , t , or any $N/4$ label. Under the identification thesis (HCSM-23), the coordinates $(2, 2)$ and $(6, 6)$ are the W boson and the top quark respectively, and the coordinate pair $\{(2, 2), (6, 6)\}$ is the observer pair $\{W, t\}$. The fixed-locus property is derived; the identification of the pair with $\{W, t\}$ is anchored.

Remark 6. Theorem 4 gives three independent characterizations of the coordinate pair $\{(2, 2), (6, 6)\}$: as the fixed locus of the chiral reflection rs , as the zero locus of the zero-state vector, and as the $\sigma = 0$ line of the charge theorem. That these three characterizations coincide is not a coincidence; it is a consequence of the fact (proved in Section 4) that rs is the unique chiral element of D_4 .

4 The Chirality Selection Theorem

This section proves that the diagonal reflection rs is the unique chiral element of D_4 in the sense of anticommuting with the winding generator. The result is derived from the substrate.

4.1 Commutation and anticommutation with H_{wind}

Definition 7 (Chiral element). An element $g \in D_4$ is called chiral if $gH_{\text{wind}}g^{-1} = -H_{\text{wind}}$, i.e., if g anticommutes with the winding generator.

Lemma 8 (Chirality of D_4 elements). *Among the five order-two elements of D_4 , only rs anticommutes with H_{wind} .*

Proof. Direct computation of the commutator $[g, H_{\text{wind}}]$ and the anticommutator $\{g, H_{\text{wind}}\}$ for each of the five involutions r^2, s, rs, r^2s, r^3s . The Frobenius norms are shown in Table 1.

Element g	$\ [g, H_{\text{wind}}]\ _F$	$\ \{g, H_{\text{wind}}\}\ _F$
r^2	1.131×10^1	1.386×10^1
s	1.131×10^1	1.386×10^1
rs	1.789×10^1	0.000×10^0
r^2s	1.131×10^1	1.386×10^1
r^3s	1.386×10^1	1.131×10^1

Table 1: Commutator and anticommutator Frobenius norms with H_{wind} . Only rs has $\|\{g, H_{\text{wind}}\}\|_F = 0$ at machine precision.

Only rs has $\|\{g, H_{\text{wind}}\}\|_F = 0$ exactly (at machine precision), and its commutator norm is nonzero. Hence rs is the unique element that anticommutes with H_{wind} . $\square$

4.2 The Chirality Selection Theorem

Theorem 9 (Chirality Selection Theorem). *The diagonal reflection $rs \in D_4$ is the unique element that (i) fixes exactly the coordinate pair $\{(2, 2), (6, 6)\}$ on the fourteen-mode multiplet, and (ii) anticommutes with the winding generator H_{wind} .*

Proof. Uniqueness follows from Lemma 1 (only rs fixes $\{(2, 2), (6, 6)\}$; the other involutions have different fixed sets) and Lemma 8 (only rs anticommutes with H_{wind}). Both properties hold for rs . $\square$

Corollary 10 (Observer pair as chiral boundary). *The coordinate pair $\{(2, 2), (6, 6)\}$ is the fixed locus of the unique chiral reflection of D_4 . Under the identification thesis (HCSM-23), the pair is $\{W, t\}$ and is the boundary through which the machine breaks chirality.*

Remark 11. The term “observer pair” has a structural content. The two members of the pair are the only coordinates on which the chiral reflection acts trivially, and they are simultaneously the only coordinates with vanishing zero-state amplitude. In a system whose machine is built to preserve chirality except for a unique boundary, that boundary is the natural structural locus of observation.

5 The D_4 -Fixed Pair Taxonomy

This section classifies the three D_4 -involution-fixed two-element coordinate subsets of the fourteen-mode multiplet. The classification of coordinate subsets is derived; the $N/4$ -ratio column of the fingerprint table is a theorem given the empirical identification (HCSM-23).

5.1 The three fixed pairs

Lemma 12 (Classification of D_4 -fixed pairs). *The fourteen-mode multiplet contains exactly three D_4 -involution-fixed two-element coordinate subsets:*

$$\{(0, 4), (4, 0)\} \quad \text{fixed by } r^2, s, r^2s, \quad (28)$$

$$\{(2, 2), (6, 6)\} \quad \text{fixed by } rs, \quad (29)$$

$$\{(2, 6), (6, 2)\} \quad \text{fixed by } r^3s. \quad (30)$$

Proof. Direct computation of the fixed set of each of the five order-two elements of D_4 . The five fixed sets are

$$\text{Fix}(r^2) = \{(0, 4), (4, 0)\}, \quad (31)$$

$$\text{Fix}(s) = \{(0, 4), (4, 0)\}, \quad (32)$$

$$\text{Fix}(rs) = \{(2, 2), (6, 6)\}, \quad (33)$$

$$\text{Fix}(r^2s) = \{(0, 4), (4, 0)\}, \quad (34)$$

$$\text{Fix}(r^3s) = \{(2, 6), (6, 2)\}. \quad (35)$$

Only three distinct coordinate subsets appear. $\square$

5.2 Structural fingerprints

Theorem 13 (D_4 -Fixed Pair Taxonomy). *Given the empirical identification (HCSM-23), the three D_4 -fixed coordinate pairs are distinguished in every structural feature of the framework: winding class, fermion/boson content, σ -line membership, and $N/4$ ratio. The fingerprints are shown in Table 2.*

Coordinate pair $N/4$ ratio	SM names	Winding class	F/B	σ -line
$\{(0, 4), (4, 0)\}$ 5/2	$\{\nu, e\}$	A, A	F, F	$-1, +1$
$\{(2, 2), (6, 6)\}$ 4	$\{W, t\}$	A, C	B, F	$0, 0$
$\{(2, 6), (6, 2)\}$ 2	$\{Z, H\}$	C, C	B, B	$-1, +1$

Table 2: The three D_4 -fixed pair fingerprints. The winding class, F/B content, and σ -line columns are derived from the substrate and the zero state. The $N/4$ ratio column is a theorem given the empirical identification (HCSM-23).

Proof. The winding-class, F/B , and σ -line columns are direct from the assignments of Section 2: $\nu, e, t \in A$; $\text{dark}, W, Z, H \in C$; $F = \{\nu, e, u, d, s, \mu, c, \tau, b, t\}$; $B = \{\text{dark}, W, Z, H\}$; $\sigma(\nu) = -1$; $\sigma(e) = +1$; $\sigma(W) = \sigma(t) = 0$; $\sigma(Z) = -1$; $\sigma(H) = +1$.

The $N/4$ -ratio column is computed from the empirical anchors (HCSM-23): $N/4(\nu) = 60$, $N/4(e) = 24$, so the ratio is $60/24 = 5/2$; $N/4(W) = 2$, $N/4(t) = 1/2$, so the ratio is $2/(1/2) = 4$; $N/4(Z) = 2$, $N/4(H) = 1$, so the ratio is $2/1 = 2$. $\square$

Corollary 14 (Diagonal classification). *The three D_4 -fixed coordinate pairs realize all three $\{A, C\}$ pairings of the winding class (AA , AC , CC) and all three $\{F, B\}$ pairings (FF , BF , BB). Given the empirical identification (HCSM-23), they also realize the three distinct $N/4$ ratios $\{(d+1)/2, d, d/2\} = \{5/2, 4, 2\}$. The coordinate pair $\{(2, 2), (6, 6)\}$ is the unique one whose members both lie on the $\sigma = 0$ line.*

Proof. The winding class and F/B pairings are read directly from Table 2. The $N/4$ ratios are the ratio column of the same table, a theorem given the anchor. The $\sigma = 0$ uniqueness is Lemma 1 applied to the σ -line partition of Section 2. $\square$

Remark 15. The class B (the size-7 middle class) is entirely absent from all three D_4 -fixed coordinate pairs. This is a structural feature of the multiplet: the middle class contains no fixed pair.

6 The $(d+1)/2$ Orbit Signature

This section proves that, given the empirical identification (HCSM-23), the framework constant $(d+1)/2 = 5/2$ manifests in each of the three D_4 orbits through a distinct binary operation.

6.1 The $N/4$ values by orbit

The $N/4$ anchors of the fourteen modes, grouped by D_4 orbit, are:

$$\mathcal{O}_0 : \{60, 24\}, \quad (36)$$

$$\mathcal{O}_1 : \{21, 20, 14, 14, 12, 9.5, 9, 7.5\}, \quad (37)$$

$$\mathcal{O}_2 : \{2, 2, 1, 0.5\}. \quad (38)$$

The coordinate pair $\{(2, 2), (6, 6)\}$ lies in $\mathcal{O}_2$.

6.2 The three operations

Theorem 16 ($(d+1)/2$ Orbit Signature). *Given the empirical identification (HCSM-23), the framework constant $(d+1)/2 = 5/2$ is realized in each of the three D_4 orbits via a distinct binary operation:*

$$\mathcal{O}_0 : \frac{N/4(\nu)}{N/4(e)} = \frac{60}{24} = \frac{5}{2} = \frac{d+1}{2}, \quad (39)$$

$$\mathcal{O}_1 : N/4(\text{dark}) - N/4(c) = 12 - 9.5 = \frac{5}{2} = \frac{d+1}{2}, \quad (40)$$

$$\mathcal{O}_2 : N/4(W) + N/4(t) = 2 + 0.5 = \frac{5}{2} = \frac{d+1}{2}. \quad (41)$$

Proof. Direct arithmetic on the $N/4$ anchors of Section 2. $\square$

6.3 Specificity of the signature

Proposition 17 (Sparsity of the $(d+1)/2$ signature). *Given the empirical identification (HCSM-23), among the 91 unordered pairs of the fourteen modes and the four binary operations (sum,*

difference, product, ratio), only four combinations hit $(d+1)/2 = 5/2$:

$$\nu/e : \text{ratio } 2.5, \quad (42)$$

$$\text{dark}/c : \text{difference } 2.5, \quad (43)$$

$$W/t : \text{sum } 2.5, \quad (44)$$

$$Z/t : \text{sum } 2.5. \quad (45)$$

The hit rate is $4/364 \approx 1.1\%$.

Proof. Exhaustive enumeration over the 91 unordered pairs and the four binary operations. $\square$

Remark 18. The Z/t duplicate of the W/t signature occurs because W and Z share the same $N/4$ value 2. The observer pair realizes the $(d+1)/2$ sum signature uniquely as a pair on the $\sigma = 0$ line, since the other candidate Z lies on the $\sigma = -1$ line. This is confirmed by the uniqueness theorem of Section 7.

7 The Observer-Pair Uniqueness Theorem

This section proves that, given the empirical identification (HCSM-23), the observer pair is the unique pair of modes satisfying five independent structural constraints simultaneously.

7.1 The five constraints

Definition 19 (Observer-pair constraints). For a pair $\{m_1, m_2\}$ of modes in the fourteen-mode multiplet, define the following constraints:

C1 (sum). $(N/4)_{m_1} + (N/4)_{m_2} = (d+1)/2$.

C2 (difference). $|(N/4)_{m_1} - (N/4)_{m_2}| = (d-1)/2$.

C3 (reflection-fixed). Both m_1 and m_2 are fixed by the diagonal reflection rs .

C4 (σ -line). Both m_1 and m_2 lie on the $\sigma = 0$ line.

C5 (fermion/boson straddling). m_1 and m_2 lie in different sectors of the F/B split.

7.2 Uniqueness

Theorem 20 (Observer-Pair Uniqueness Theorem). *Given the empirical identification (HCSM-23), the pair $\{W, t\}$ is the unique pair of modes in the fourteen-mode multiplet satisfying all five constraints C1–C5 simultaneously. The uniqueness is a uniqueness within the family of anchors: it is a structural relation among the $N/4$ values, not a derivation of which mode is W or which is t .*

Proof. We proceed in three steps.

Step 1: C1 and C2 alone determine the numeric values. The linear system

$$x + y = \frac{d+1}{2} = \frac{5}{2}, \quad |x - y| = \frac{d-1}{2} = \frac{3}{2} \quad (46)$$

has solutions $(x, y) = (2, 1/2)$ or $(1/2, 2)$. Both solutions are realized among the $N/4$ anchors: $\{2, 1/2\}$ appears as $\{N/4(W), N/4(t)\}$ and as $\{N/4(Z), N/4(t)\}$ (since W and Z share the value $N/4 = 2$).

Step 2: C3 and C4 eliminate the $\{Z, t\}$ candidate. The pair $\{Z, t\}$ fails C3, because $rs(Z) = H \neq Z$, and fails C4, because $\sigma(Z) = -1 \neq 0$. Hence $\{Z, t\}$ is eliminated.

Step 3: no other pair satisfies even three of the constraints. Exhaustive enumeration confirms that no pair other than $\{W, t\}$ and $\{Z, t\}$ satisfies both C1 and C2, and that $\{W, t\}$ is the unique pair satisfying both C1, C2, C3, C4. Constraint C5 is automatically satisfied by $\{W, t\}$ (since $W \in B$ and $t \in F$) and by $\{Z, t\}$ (since $Z \in B$ and $t \in F$), but the latter is already eliminated. $\square$

Corollary 21 (Solution of the linear system). *Given the empirical identification (HCSM-23), the pair $\{W, t\}$ is the unique solution to the two-by-two linear system*

$$N/4(W) + N/4(t) = \frac{d+1}{2}, \quad N/4(W) - N/4(t) = \frac{d-1}{2}, \quad (47)$$

with solution

$$N/4(W) = \frac{d}{2}, \quad N/4(t) = \frac{2}{d}. \quad (48)$$

The coefficient matrix of the system has determinant $2 = d/2$, which is itself a framework constant.

Proof. Solving the linear system gives $N/4(W) = ((d+1)/2 + (d-1)/2)/2 = d/2$ and $N/4(t) = ((d+1)/2 - (d-1)/2)/2 = 2/d$. The determinant is $1 \cdot 1 - 1 \cdot (-1) = 2$. At $d = 4$, this gives $N/4(W) = 2$ and $N/4(t) = 1/2$, matching the anchors. $\square$

Remark 22. The two constraints C1 and C2 have independent structural meanings, given the identification. C1 states that the observer pair's $N/4$ sum equals the coefficient $(d+1)/2 = |F|/|B|$. C2 states that the pair's $N/4$ separation equals $(d-1)/2$, the same constant that appears in the kL crossover identity of HCSM-51 (Section 9). Together they determine the pair's $N/4$ values uniquely within the anchor family.

8 The Swap-Pair Fingerprint Theorem

This section proves that, given the empirical identification (HCSM-23), the orbit-class swap pair $\{\text{dark}, t\}$ is the unique two-element subset of the fourteen-mode multiplet whose four symmetric-function values are all framework constants.

8.1 The orbit-class swap

The D_4 orbit partition and the winding class partition of the fourteen-mode multiplet differ by exactly one two-element coordinate swap.

Lemma 23 (Orbit-class reconstruction). *The winding class partition $\{A, B, C\}$ is the D_4 orbit partition $\{\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2\}$ modified by the unique two-mode swap*

$$t : \mathcal{O}_2 \rightarrow A, \quad \text{dark} : \mathcal{O}_1 \rightarrow C. \quad (49)$$

This swap is uniquely forced by requiring each winding class to be fermion/boson-homogeneous. Under the identification thesis (HCSM-23), the swapped coordinates correspond to the top quark t and the dark photon.

Proof. The orbit partition and the class partition differ only on the two coordinates (6,6) and (5,7): (6,6) is in $\mathcal{O}_2$ but in class A ; (5,7) is in $\mathcal{O}_1$ but in class C . All other coordinates lie in the same orbit and the same class.

Without the swap, the classes are not fermion/boson-homogeneous: the orbit $\mathcal{O}_2$ contains three bosons $\{(2, 2), (2, 6), (6, 2)\}$ and one fermion $\{(6, 6)\}$, so it is not F/B -homogeneous; the orbit $\mathcal{O}_1$ contains one boson $\{(5, 7)\}$ and seven fermions, so it is not F/B -homogeneous either. Removing $(6, 6)$ from $\mathcal{O}_2$ and adding $(5, 7)$ to make class C gives $C = \{(5, 7), (2, 2), (2, 6), (6, 2)\}$ (all bosons), $A = \{(0, 4), (4, 0), (6, 6)\}$ (all fermions), and $B = \mathcal{O}_1 \setminus \{(5, 7)\}$ (all fermions). Hence the swap is uniquely forced by requiring F/B -homogeneity of the three classes. $\square$

8.2 The symmetric-function fingerprint

Definition 24 (Symmetric functions of a pair). For a pair $\{a, b\}$ of modes with $N/4$ values x and y , define the four symmetric functions

$$\text{sum}(a, b) = x + y, \quad \text{diff}(a, b) = |x - y|, \quad \text{prod}(a, b) = x \cdot y, \quad \text{ratio}(a, b) = \frac{\max(x, y)}{\min(x, y)}. \quad (50)$$

Theorem 25 (Swap-Pair Fingerprint Theorem). *Given the empirical identification (HCSM-23), the swap pair $\{\text{dark}, t\}$ is the unique two-element subset of the fourteen-mode multiplet whose four symmetric-function values are simultaneously all framework constants:*

$$\text{sum}(\text{dark}, t) = 12 + \frac{1}{2} = \frac{25}{2} = \frac{(d+1)^2}{2}, \quad (51)$$

$$\text{diff}(\text{dark}, t) = 12 - \frac{1}{2} = \frac{23}{2} = \frac{H+2d-1}{2}, \quad (52)$$

$$\text{prod}(\text{dark}, t) = 12 \cdot \frac{1}{2} = 6 = 2(d-1), \quad (53)$$

$$\text{ratio}(\text{dark}, t) = \frac{12}{1/2} = 24 = d! = N/4(e). \quad (54)$$

Proof. Direct arithmetic on the anchors $N/4(\text{dark}) = 12$ and $N/4(t) = 1/2$: $\text{sum} = 12.5$, $\text{diff} = 11.5$, $\text{prod} = 6$, $\text{ratio} = 24$.

Uniqueness is established by exhaustive enumeration of all $\binom{14}{2} = 91$ unordered pairs and comparison of their four symmetric-function values. The results are tabulated in Table 3. No pair other than $\{\text{dark}, t\}$ matches all four framework-constant values simultaneously. $\square$

Symmetric function	Framework-constant value	Pairs hitting it	Pairs hitting all four
sum	$25/2 = 12.5$	34	1
diff	$23/2 = 11.5$	5	1
prod	$2(d-1) = 6$	4	1
ratio	$d! = 24$	1	1

Table 3: Uniqueness of the swap-pair fingerprint. Only $\{\text{dark}, t\}$ hits all four.

Corollary 26 (Framework-constant sector bridge). *The four constants appearing in the swap-pair fingerprint are drawn from four distinct sectors of the framework:*

- $25 = (d+1)^2$: the Newton-constant sector (the bilinear trace of the hidden B_1 isotypic sector; see HCSM-48 [12]).
- $23 = H + 2d - 1$: the principal-angle sector (the trace numerator of the loop-commutant Gram matrix; see HCSM-17 [7]).

- $2(d-1) = 6$: the dimensional-rigidity sector (the offset $2(d-1)$; see HCSM-51 [13]).
- $24 = d! = N/4(e)$: the mass sector (the electron $N/4$ anchor).

The swap pair therefore provides a bridge between these four sectors.

Remark 27. The appearance of 24 in the ratio slot is notable: it equals both the factorial $d! = 4! = 24$ and the electron's $N/4$ anchor $N/4(e) = 24$, and the framework constant $V/2 = 48/2 = 24$. The three identifications are equivalent at $d = 4$.

9 The Dimensional-Rigidity Statement

This section states the dimensional-rigidity content linking the observer pair to the kL crossover. The statement comprises two independent constraints on d and one definitional evaluation at $d = 4$. Each component is labeled with its derivation status.

9.1 The two constraints on d

Theorem 28 (Dimensional-rigidity constraints). *The following two independent identities on d each have the unique positive integer solution $d = 4$:*

$$(observer-pair\ separation) \quad \frac{d}{2} - \frac{2}{d} = \frac{d-1}{2}, \quad (55)$$

$$(kL\ crossover, HCSM-51) \quad (d-1)^2 = 2d+1. \quad (56)$$

Both reduce to the polynomial equation $d^2 - 4d = 0$.

Proof. For the first identity: $\frac{d}{2} - \frac{2}{d} = \frac{d^2-4}{2d}$. Setting this equal to $\frac{d-1}{2}$ gives $d^2 - 4 = d^2 - d$, i.e., $-4 = -d$, i.e., $d = 4$. Equivalently, $d^2 - 4d = d(d-4) = 0$, which has solutions $d \in \{0, 4\}$; the positive integer solution is $d = 4$. The left-hand side is a function of the observer-pair anchor values $N/4(W) = d/2$ and $N/4(t) = 2/d$ (Corollary 21), so the identity is a theorem given the empirical identification (HCSM-23).

For the second identity: $(d-1)^2 = 2d+1$ expands to $d^2 - 2d + 1 = 2d + 1$, i.e., $d^2 - 4d = 0$, i.e., $d(d-4) = 0$. This identity is derived in HCSM-51 [13] from the bare kL crossover, independently of the $N/4$ anchors. $\square$

Remark 29. The two identities are independent as equations, but they share the same solution set. That both framework relations select the same value $d = 4$ is a nontrivial consistency check on the framework: the observer-pair anchor values and the kL crossover condition are derived from different parts of the framework, yet agree on the physical dimension.

9.2 The dark energy coefficient identity

Theorem 30 (Observer-pair sum equals dark energy coefficient). *Given the empirical identification (HCSM-23), the observer pair's $N/4$ sum equals the dark energy coefficient and the fermion-to-boson multiplicity ratio:*

$$N/4(W) + N/4(t) = \frac{d+1}{2} = \frac{|F|}{|B|} = \frac{\rho_{DE}}{m_\nu^4}. \quad (57)$$

Proof. Direct substitution: $N/4(W) + N/4(t) = 2 + 1/2 = 5/2$; $|F|/|B| = 10/4 = 5/2$; $(d+1)/2 = 5/2$ at $d = 4$. $\square$

Remark 31. The equality $|F|/|B| = (d+1)/2$ is a definitional identity given the F/B split (from HCSM-13: $|F| = d(d+1)/2$ and $|B| = d$). The equality with the observer pair's $N/4$ sum is a theorem given the empirical identification. The two equalities are not independent constraints on d ; both evaluate to $5/2$ at $d = 4$. The dark energy coefficient $(d+1)/2$ is derived in HCSM-53 [14]; the observer-pair sum is the same value at the anchor level.

9.3 Summary of the dimensional-rigidity content

The dimensional-rigidity content of this section comprises:

- (1) A theorem given the anchor: the observer-pair separation identity $\frac{d}{2} - \frac{2}{d} = \frac{d-1}{2}$ (identity 1 of Theorem 28), with unique positive integer solution $d = 4$.
- (2) A derived theorem (HCSM-51): the kL crossover identity $(d-1)^2 = 2d+1$ (identity 2 of Theorem 28), also with unique positive integer solution $d = 4$.
- (3) A definitional evaluation at $d = 4$: the equality $N/4(W) + N/4(t) = |F|/|B| = (d+1)/2 = 5/2$ (Theorem 30), which shares the value $5/2$ with the observer pair's sum but does not independently constrain d .

Identities 1 and 2 are independent constraints on d ; both select the same physical dimension. Identity 3 is a value coincidence at that dimension.

10 Discussion

10.1 What the observer pair is

The coordinate pair $\{(2,2), (6,6)\}$ is the fixed locus of the unique chiral reflection of D_4 , the zero locus of the zero-state vector, and the $\sigma = 0$ line of the charge theorem. It is the unique reflection-fixed coordinate pair straddling the fermion/boson boundary. Under the identification thesis (HCSM-23), the pair is the W boson and the top quark, and it is the boundary through which the machine breaks its own chirality.

The two members of the pair are the two Standard Model particles most directly tied to the electroweak scale: the W boson is the charged electroweak boson, and the top quark is the heaviest fermion and the principal contributor to the Higgs vacuum expectation value through its Yukawa coupling. The framework's structural prediction, conditional on the empirical identification, is that they are paired as observers of the machine.

10.2 The three structural regions

The fourteen-mode multiplet partitions into three structurally distinct regions. The classification of the regions as Klein orbits is from HCSM-27 [11]; the reader is referred there for the full derivation.

- The four two-cycle coordinate orbits with kernel $\{e, rs \cdot T_{\text{half}}\}$, which are homogeneous in orbit, class, and F/B , and heterogeneous in σ .
- The observer coordinate orbit $\{(2,2), (6,6)\}$, which is the unique orbit with kernel $\{e, rs\}$.
- The four-cycle orbit $\{(5,7), (3,1), (1,3), (7,5)\}$, which is the unique orbit with trivial kernel and contains the swap partner of the top quark coordinate.

The observer coordinate pair occupies the middle region. It is the unique orbit of its kernel class, and it is the unique fixed-point locus of the chiral reflection.

10.3 The role of the swap

The orbit-class swap pair $\{(5, 7), (6, 6)\}$ connects the four-cycle orbit to the observer orbit. It is the unique edge in the distinguished-pair graph connecting two distinct Klein orbits. Under the identification thesis, this swap pair is $\{\text{dark}, t\}$. Its four-symmetric-function fingerprint (Theorem 25) is entirely framework-constant, drawing on the Newton-constant sector, the principal-angle sector, the dimensional-rigidity sector, and the mass sector. This is the framework's structural bridge between the observer pair and the rest of the fourteen-mode multiplet.

11 Inputs and derived

11.1 Derived from the substrate

The following results are proved without reference to the $N/4$ anchors or to the particle identifications:

- (1) The fixed set of rs is the coordinate pair $\{(2, 2), (6, 6)\}$ (Lemma 1).
- (2) The zero-state vector is odd under rs (Lemma 2).
- (3) The zero-state amplitude vanishes on the fixed set (Corollary 3).
- (4) The three characterizations of $\{(2, 2), (6, 6)\}$ as $\text{Fix}(rs)$, the zero locus of s , and the $\sigma = 0$ line coincide (Theorem 4).
- (5) Among the five involutions of D_4 , only rs anticommutes with H_{wind} (Lemma 8).
- (6) rs is the unique element of D_4 that fixes $\{(2, 2), (6, 6)\}$ and anticommutes with H_{wind} (Theorem 9).
- (7) The fourteen-mode multiplet contains exactly three D_4 -involution-fixed coordinate pairs (Lemma 12).
- (8) The winding class partition is the D_4 orbit partition modified by the unique two-coordinate swap that makes each class fermion/boson-homogeneous (Lemma 23).
- (9) The kL crossover identity $(d - 1)^2 = 2d + 1$ (HCSM-51, cited in Theorem 28).

11.2 Theorems given the empirical identification (HCSM-23)

The following results are theorems about the structural relations among the $N/4$ anchors:

- (10) The $N/4$ -ratio fingerprint of the three D_4 -fixed pairs (Theorem 13).
- (11) The $N/4$ -ratio classification of the three fixed pairs (Corollary 14).
- (12) The $(d + 1)/2$ orbit signature (Theorem 16).
- (13) The sparsity of the $(d + 1)/2$ signature (Proposition 17).
- (14) The observer-pair uniqueness under the five constraints (Theorem 20).
- (15) The solution of the observer-pair linear system (Corollary 21).
- (16) The swap-pair fingerprint (Theorem 25).

- (17) The framework-constant sector bridge (Corollary 26).
- (18) The observer-pair separation identity (identity 1 of Theorem 28).
- (19) The observer-pair sum equals the dark energy coefficient (Theorem 30).

11.3 Inputs

- (1) **Input 1 (SM content).** The Standard Model particles and their observed properties, as used to identify the fourteen modes.
- (2) **Input 2 (physical identifications).** The identifications $A_1 \leftrightarrow \text{color}$, $B_1 \leftrightarrow \text{weakisospin}$.
- (3) **Input 3 (the identification thesis, HCSM-23).** The assignment of each of the fourteen modes at $\lambda = 4$ to a specific Standard Model particle, together with the $N/4$ mass label attached to each. This is the framework's empirical anchor, structurally parallel to the Standard Model's Yukawa couplings.
- (4) **The framework constants** $d = 4$, $N = 64$, $L = 8$, $H = 16$ (HCSM-01 [2]).

11.4 What is not claimed

We do not claim to derive the identification thesis (HCSM-23) from the substrate. The assignment of the fourteen modes to the fourteen particles, and the $N/4$ labels attached to each, are the framework's empirical anchors.

We do not claim that the four theorems of Section 6–8 are HCSM-native without qualification. They are theorems given the anchor: they hold for the specific anchor values observed in the Standard Model, and they are falsifiable if the anchor values differ.

We do not claim that the naming of the coordinates $(2, 2)$ and $(6, 6)$ as W and t is derived. The coordinates and the fixed-locus property are derived; the naming is anchored.

We do not claim that the dark energy coefficient identity (Theorem 30) is an independent constraint on d . It is a value coincidence at $d = 4$ between two framework quantities that share the same algebraic expression.

The framework's claim is that it has fewer empirical anchors than the Standard Model, not that it has none.

12 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-26_verification`. The script constructs the fourteen-mode multiplet, the D_4 action, the winding generator, the zero-state vector, the $N/4$ anchors, and the charge functional from HCSM-native definitions, and verifies:

- (1) The fixed set of each order-two element of D_4 on the fourteen-mode multiplet, using coordinates only (no $N/4$ anchors).
- (2) The oddness of s under rs and the vanishing of s on the fixed set, using coordinates only.
- (3) The commutation/anticommutation structure of each D_4 element with H_{wind} , using the matrix H_{wind} only.

- (4) The coordinate-level D_4 -fixed pair classification and its F/B and σ -line fingerprints.
- (5) The $N/4$ -ratio fingerprint of the three fixed pairs (given the anchors).
- (6) The $(d+1)/2$ orbit signature and its sparsity (given the anchors).
- (7) The observer-pair uniqueness under the five constraints (given the anchors).
- (8) The swap-pair fingerprint uniqueness, all 91 pairs enumerated (given the anchors).
- (9) The two dimensional-rigidity constraints of Theorem 28 and the value coincidence of Theorem 30.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

13 Conclusion

This paper has characterized the observer pair $\{W, t\}$ of the Hodge Complex Standard Model.

Two theorems are fully derived from the substrate and do not use the $N/4$ anchors. The Fixed Locus Theorem establishes that the coordinate pair $\{(2, 2), (6, 6)\}$ is the fixed locus of the diagonal reflection rs of D_4 , coincides with the zero locus of the zero-state vector, and is the $\sigma = 0$ line of the charge theorem. The Chirality Selection Theorem establishes that rs is the unique element of D_4 that both fixes $\{(2, 2), (6, 6)\}$ and anticommutes with the winding generator.

Four theorems are conditional on the empirical identification thesis (HCSM-23). Given the anchor, the pair's $N/4$ labels are the unique solution to the linear system determined by $(d+1)/2$ and $(d-1)/2$; the framework constant $(d+1)/2$ manifests in each of the three D_4 orbits through a distinct binary operation, and the observer pair realizes the sum signature; the orbit-class swap pair $\{\text{dark}, t\}$ is the unique two-element subset whose four symmetric-function values are simultaneously all framework constants. These theorems are falsifiable: they hold for the specific anchor values observed in the Standard Model.

The dimensional-rigidity statement comprises two independent constraints on d (the observer-pair separation identity and the kL crossover identity) each of which uniquely selects $d = 4$, plus one definitional evaluation at $d = 4$ in which two framework quantities share the value $5/2$.

The framework's claim is that the Standard Model is what the substrate looks like when you observe it. The identification of the fourteen modes with the fourteen particles is the empirical thesis; the structural relations among the $N/4$ anchors are theorems given the thesis. The framework's strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

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DC.subject: Hodge Complex Standard Model; HCSM; observer pair; W boson; top quark; D_4 representation theory; fixed locus; chiral reflection; identification thesis; $N/4$ anchors; dimensional rigidity; dark energy coefficient

DC.description: The Hodge Complex Standard Model assigns the fourteen modes at $\lambda = 4$ to one Standard Model generation. The identification of the modes with specific particles, and the $N/4$ mass labels, are the framework’s empirical thesis (HCSM-23). This paper characterizes the coordinate pair $\{(2, 2), (6, 6)\}$. Two theorems are fully derived from the substrate: the Fixed-Locus Theorem and the Chirality Selection Theorem. Four theorems are conditional on the empirical identification: the D_4 -fixed pair taxonomy, the $(d + 1)/2$ orbit signature, the observer-pair uniqueness theorem, and the swap-pair fingerprint theorem. The dimensional-rigidity statement comprises two independent constraints on d , each selecting $d = 4$, plus one definitional evaluation at $d = 4$.

DC.identifier: HCSM-26-2026-09

A.2 BibTeX for Automated Indexing

```
@article{PreschuttiHCSM26,
  author = {Preschutti, Stanley},
  title = {{HCSM-26}: The Observer Pair  $\{W, t\}$ },
  journal = {HCSM White Paper Series},
  year = {2026},
  month = {September},
  note = {ORCID: 0009-0004-5445-1744},
  keywords = {Hodge Complex Standard Model, HCSM, observer pair,
    W boson, top quark, D4 representation theory, fixed locus,
    chiral reflection, identification thesis, N/4 anchors,
    dimensional rigidity},
  abstract = {The coordinate pair  $\{(2,2), (6,6)\}$  of the HCSM
    fourteen-mode multiplet is the fixed locus of the unique
    chiral reflection of D4 (fully derived). Given the empirical
    identification thesis (HCSM-23), the pair is  $\{W, t\}$  and
    exhibits four additional structural theorems: the D4-fixed
    pair taxonomy, the  $(d+1)/2$  orbit signature, the
    observer-pair uniqueness theorem, and the swap-pair
    fingerprint theorem.}
}
```